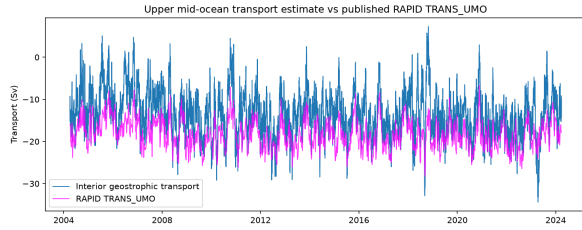


Geostrophic transport and relating AMOC series

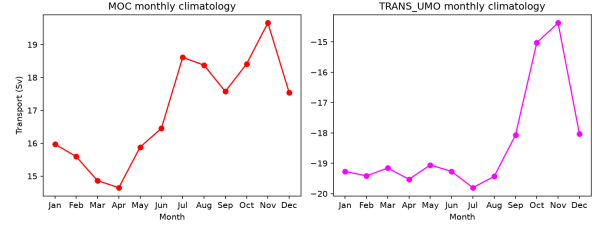
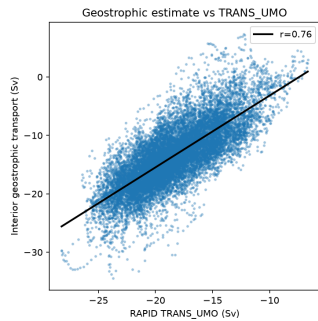
<https://github.com/Rosecodee/assignment2>

Part 1 — Geostrophic transport

The Upper Meridional Overturning (UMO) transport was calculated using TEOS-10 dynamic height differences between the eastern and western boundaries, vertically integrated to the average AMOC maximum depth (1100 m). The integration was deliberately restricted to these upper 1100 m rather than the full ocean depth to isolate the southward-flowing interior transport, which, together with the northward-flowing Ekman and Florida Current transports, defines the upper limb of the AMOC. Extending the integration beyond this depth would merge opposing flow regimes with the deep boundary currents and obscure the upper-ocean return flow structure.



The interior geostrophic estimate displays strong agreement with the published TRANS UMO, yielding a linear correlation coefficient of $r = 0.76$. While the signals match closely, the geostrophic estimate exhibits higher variability (broader dynamic range from -35 Sv to $+5$ Sv) compared to TRANS UMO.



Part 2 — Evaluating time series

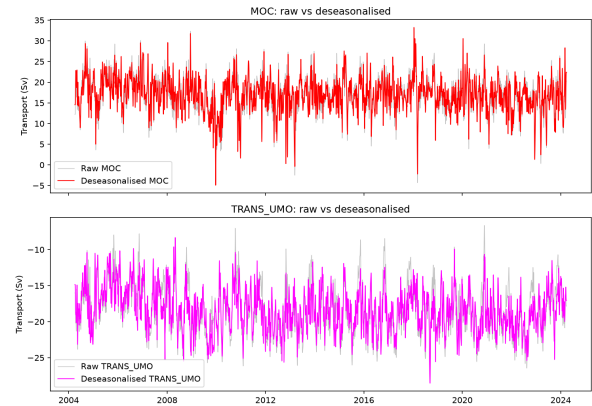
2A — Single series: seasonal cycle and trend

The monthly climatology, deseasonalised series, and autocorrelation were computed for both MOC and TRANS UMO using the `remove_seasonal_cycle`, `autocorr`, and `integral_timescale` functions implemented in `correlation_trends`.

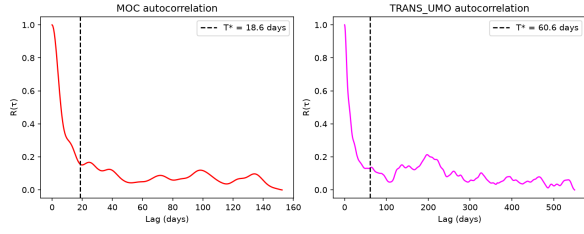
Both 26°N MOC and TRANS UMO exhibit distinct seasonal climatologies:

MOC: Reaches its minimum transport in April (~ 14.6 Sv) and peaks in November (~ 19.8 Sv).

TRANS UMO: Shows a minimum southward transport (least negative) in November (~ -14.3 Sv) and maximum southward flow in July (~ -19.8 Sv).



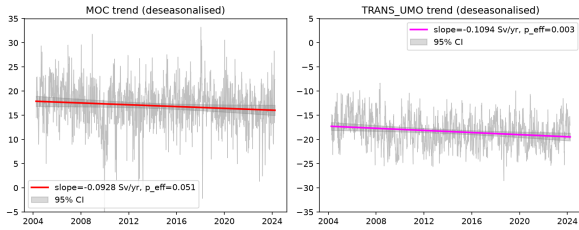
Removing the monthly climatologies yields deseasonalised time series that preserve their respective mean levels (≈ 17.2 Sv for MOC and ≈ -17.5 Sv for TRANS UMO).



The integral decorrelation timescale (T^*) was evaluated to determine the effective sample size ($EDOF = N_{\text{raw}}/(2T^*)$):

MOC: $T^* = 18.6$ days, indicating fast atmospheric/Ekman-driven decorrelation.

TRANS UMO: $T^* = 60.6$ days, reflecting the longer baroclinic/mesoscale oceanic memory of the interior field.



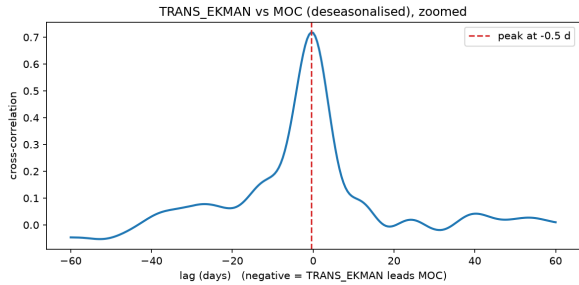
Linear trends were fitted to the deseasonalised series, evaluating significance via effective degrees of freedom (p_{eff}):

MOC Trend: -0.0928 ± 0.047 Sv/yr. With $p_{\text{eff}} = 0.051$, the trend is not statistically significant at the 95% confidence level ($p > 0.05$), highlighting that ignoring autocorrelation would have falsely suggested a significant decline.

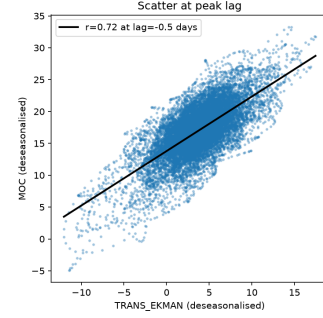
TRANS UMO Trend: -0.1094 ± 0.035 Sv/yr. With $p_{\text{eff}} = 0.003$, this strengthening southward transport trend remains statistically significant ($p < 0.05$) even after accounting for high temporal persistence.

2B — A pair of series: cross-correlation and lead/lag

Both TRANS EKMAN and MOC are derived from the same standardised RAPID product and are defined on an identical TIME grid. Therefore, no additional processing was required to match the filtering or align the two series to a common time grid.



The -0.5 day lag demonstrates that surface wind-stress-driven Ekman transport leads the overall overturning transport almost instantaneously. Because Ekman transport contributes directly to the upper-limb transport budget, fluctuations in wind stress immediately project onto the total MOC. Given the short memory of $T^* \approx 18.6$ days, the relationship provides strong physical evidence that short-term AMOC variability at 26°N is heavily wind-driven rather than density-driven.



However, this near-zero-lag relationship should not be interpreted as purely dynamical, since by construction $\text{MOC} = \text{TRANS FC} + \text{TRANS EKMAN} + \text{TRANS UMO}$. Thus, TRANS EKMAN is an additive component of MOC rather than an independent predictor. As a cross-check, the correlation between TRANS_UMO and MOC was also computed, yielding $r \approx 0.25$ at a peak lag of ~ 6.5 days, consistent with the lecture reference value of ≈ 0.28 . This confirms the analysis pipeline and indicates that the geostrophic component has a substantially weaker relationship with MOC variability than the wind-driven Ekman component.

Part 3 — Depth Sensitivity of Transport

As the additional analysis, the Part 1 interior geostrophic transport was recomputed at three integration depths (700, 1000, 1100 m) and compared both to each other and to TRANS UMO.

Table 1: Transport statistics for different maximum integration depths.

z_{max} (m)	Mean (Sv)	Std (Sv)	r vs UMO
700	-16.38	4.40	0.787
1000	-14.11	5.29	0.772
1100	-13.46	5.50	0.762

The interior geostrophic transport is clearly sensitive to the chosen integration depth. Extending the integration from 700 m to 1100 m weakens the mean southward transport (from -16.4 to -13.5 Sv) while slightly increasing its standard deviation from 4.4 Sv to 5.5 Sv reflecting the

additional, more variable water-mass contributions incorporated at depth. Correlation with the published TRANS UMO is highest at 700 m ($r = 0.79$) and decreases slightly toward 1100 m ($r = 0.76$)